GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain

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Abstract

- Addresses costly and non-transparent traditional agricultural equipment rental systems.
- Introduces GreenEquip, a blockchain-based peer-to-peer equipment rental platform.
- Uses Ethereum smart contracts and MetaMask for secure listings, bookings, and payments.
- Records all rental transactions permanently on the blockchain.
- Enables direct interaction between farmers and equipment owners.

Problem Statement and Objectives

Problem Statement

High equipment costs and inefficient, non-transparent rental systems limit farmers' direct access to machinery and lead to underutilized resources. GreenEquip proposes a blockchain-based platform for secure and transparent equipment sharing. It utilizes blockchain technology, administrative verification, and AI-driven validation and recommendations to ensure efficiency and reliability.

Objectives

- Facilitate peer-to-peer farming equipment rentals using blockchain technology.
- Automate equipment listings, bookings, and payments via smart contracts.
- Eliminate centralized middlemen to increase affordability for smallholder farmers.
- Provide transparent dashboards for real-time tracking of rental statuses.
- Ensure secure and immutable transaction records using the Ethereum network.

System Design

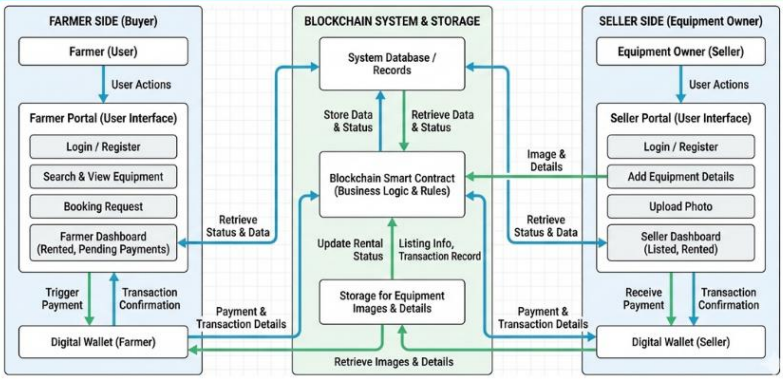

Fig 1. Block Diagram of the System

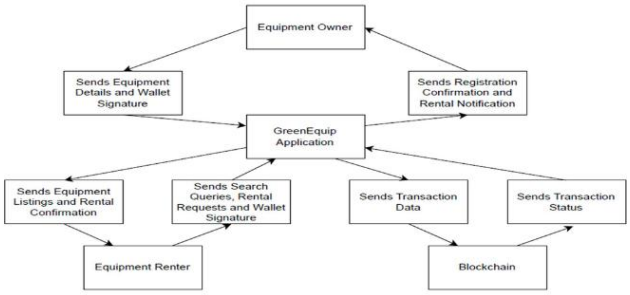

Fig 2. Diagram representing the relationship between the owner, renter, and use of blockchain technology

Results

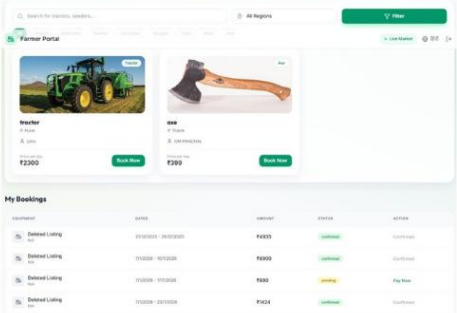

Fig 3. Farmer Side (Buyer) Portal

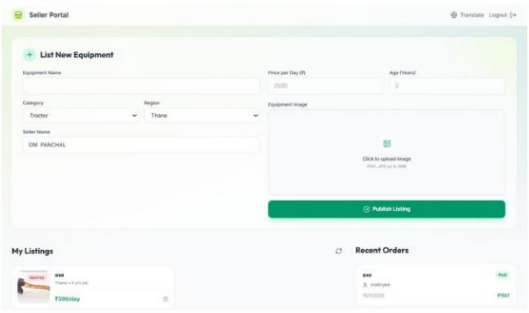

Fig. 4. Equipment Owner Side (Seller) Portal

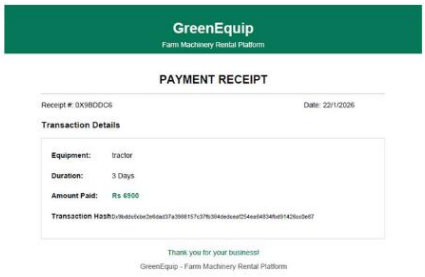

Fig. 5. Payment Receipt Generated After Transaction

SDG Goal

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Conclusion

- Demonstrated a decentralized agricultural rental model using blockchain technology.
- Automated listings, bookings, and payments through self-executing smart contracts.
- Ensured data integrity and prevented double bookings via an immutable ledger.
- Provided full traceability and accountability for both farmers and equipment owners.
- Validated a trustless, peer-to-peer architecture as a scalable solution for farming.

References

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[2] Dong, S., Abbas, K., Li, M., & Kamruzzaman J, "SP2F: Secure Privacy-Preserving Framework blending Deep Learning & Blockchain for Smart Farming", PeerJ Computer Science, 9, e1705, 2023.

[3] Omkar Jadhav, Chaitanya Gamane, Shraddha Jagtap, Paras Mahale, "Farming Equipment Rental System", International Research Journal of Modernization in Engineering Technology and Science, Volume: 07/Issue:03/March-2025.